

# Explainable Graph-Based Framework for Twitter Bot Detection

*Dr.S.Subhashini*

*Assistant Professor (Sr.Gr)/CSE*

*B.S Abdur Rahman Crescent Institute of  
Science and Technology  
Chennai, India*

[subhashini@crescent.education](mailto:subhashini@crescent.education)

*Abdul rahim S*

*Student /CSE*

*B.S Abdur Rahman Crescent Institute of  
Science and Technology  
Chennai, India*

[abdulrahim82436@gmail.com](mailto:abdulrahim82436@gmail.com)

*Ajay M*

*Student /CSE*

*B.S Abdur Rahman Crescent Institute of  
Science and Technology  
Chennai, India*

[ajayvym2025@gmail.com](mailto:ajayvym2025@gmail.com)

**Abstract** - The rise of automated accounts, or bots, on Twitter presents a serious challenge to platform integrity and public discourse. This research introduces an explainable, graph-based framework for detecting bot accounts by combining structured behavioural feature engineering with GraphSAGE node embeddings. The system analyses user profile metadata and activity-based metrics, engineering features such as posting frequency, follower spike indicators, and engagement ratios, then performs a k-nearest neighbour lookup over a training reference graph to generate a 64-dimensional embedding vector capturing each account's social context. A Random Forest classifier trained on the combined feature set achieves an AUC of 0.9737 and an accuracy of 94.53%, with an empirically derived classification threshold of 0.39 determined through precision-recall analysis. To ensure transparency, LIME is applied to generate per-account feature-level explanations, and a large language model converts these into plain-language narrative summaries accessible to non-technical stakeholders. The complete system is deployed as an interactive web application supporting on-demand account analysis with real-time classification results, LIME evidence, and AI-generated explanations. This is a valuable tool for researchers and platform administrators investigating automated account behaviour on Twitter.

**Keywords** — Twitter Bot Detection, Graph Neural Networks, Similarity Graph Construction, k-Nearest Neighbor Cosine Similarity, GraphSAGE, Node Embeddings, Feature Normalization, Bot-Human Classification, LIME Explainability, Dataset-Independent Detection Pipeline

## I. INTRODUCTION

Twitter's dominance in today's world is undeniable. It's the hub for how we communicate, get our news, and discuss important issues, connecting billions globally and allowing for instant sharing of thoughts and content. But this widespread connection also opens doors for automated accounts, or "bots." These bots vary widely, from helpful automation tools to malicious actors designed to spread false information, boost trends artificially, sway public opinion, and launch coordinated attacks. The sophistication of these bots has skyrocketed in recent years. Advanced bots now use machine learning to create human-like content, copy human behavior, and

blend seamlessly into social networks, making old detection methods less effective. Early attempts at spotting bots used manual analysis, simple rules, or basic machine learning models, focusing on user details, how often they post, and other account-specific data. However, these methods struggle to keep up with evolving bot tactics, miss coordinated actions, and often lack clear explanations. To combat these challenges, this project introduces a detailed, understandable, and graph-based system. It combines various machine learning techniques, reduces data complexity through feature selection, uses Graph Neural Networks (GNNs) like GraphSAGE to understand user similarity within the network, and employs ensemble classifiers such as Random Forest for precise bot identification. The system carefully prepares and analyzes user and content data, including activity levels, language patterns, engagement, and relationship data, to identify both individual and coordinated bot behaviors. Beyond accurate detection, the system prioritizes transparency using Explainable AI (XAI) methods like LIME. This allows analysts and platform administrators to understand the key behavioral, linguistic, and relational factors that influence the model's predictions.

## II. LITERATURE REVIEW

Research on social network bot detection (SNBD) has evolved significantly over time, with researchers exploring a wide range of techniques from traditional machine learning to more advanced deep learning models. Early studies mainly relied on simpler methods, using features like follower-following ratios, posting activity, and statistical patterns (such as Benford's law) to distinguish between genuine users and bots. These approaches proved useful in identifying basic bot behaviors and laid the groundwork for future developments [10].

As research progressed, scholars began examining the broader differences between bots and human users. Large-scale studies revealed clear distinctions in how bots and humans communicate, structure their networks, and interact online. These insights helped improve feature selection and made detection systems more effective [3]. At the same time, some researchers explored ways to involve humans in the detection process. They found that when users were supported with algorithm-generated bot scores and brief training, their ability to identify bots

improved significantly, although simple warning messages alone were not very effective [6].

With advancements in deep learning, more sophisticated models began to emerge. Researchers introduced hybrid systems that combine user profile data with behavioral patterns, achieving high accuracy even when dealing with imbalanced datasets [2]. Transformer-based models and hybrid architectures like BiGRU-LSTM further improved detection by analyzing the meaning and context of text, rather than relying only on surface-level features [4], [5].

Another important trend in the literature is the use of multiple types of features together. Combining textual data, user behavior, timing patterns, and network structure has been shown to improve the robustness of detection systems. Reviews of existing work suggest that deep learning models—especially those combining sequential and graph-based techniques—generally perform better than traditional approaches when sufficient data is available [7], [11]. In particular, network-based methods that focus on relationships between users have proven effective, especially when textual data is limited or unreliable [9]. Additionally, some systems integrate bot detection into larger frameworks, such as credibility analysis, to improve the overall trustworthiness of online content [8].

Despite these improvements, one major challenge remains: bots are constantly evolving. Models trained on older datasets often struggle to detect newer types of bots with different behaviors. This has led researchers to focus on developing more adaptive and explainable systems that can continuously learn and adjust to new patterns [1].

Overall, the field is moving away from static, rule-based approaches toward more flexible, intelligent systems that can keep up with the rapidly changing nature of social media bots.

### III. Proposed System

The proposed system aims to spot fake accounts, that is bots, on Twitter. It's designed to be accurate, and importantly, to explain why it flags an account as a bot. The system uses a combination of two approaches. By looking at individual accounts It examines details about each user, like their profile information and how they behave and looking at connections between accounts, it constructs a similarity-based network where users are linked based on behavioural and profile feature similarity, creating a graph-based representations. By combining these two perspectives, the system can identify both individual bots and coordinated bot networks. It uses a machine learning model to make its predictions, ensuring it works well even as bots become more sophisticated. The system is also designed to be easy to understand and to handle large amounts of data.

#### Proposed algorithm

##### Input:

Twitter account data features.

##### Output:

Account classification Whether Human or Bot with Explanation.

Step 1 – Start

Step 2 – Collect Twitter account data.

Step 3 – Clean the dataset, handle missing values, normalize features, and separate labels (bot or human).

Step 4 – Extract numerical features and other behavioral attributes.

Step 5 – Build a similarity graph using a K-Nearest Neighbors method.

Step 6 – Apply the GraphSAGE model to the similarity graph to learn node embeddings.

Step 7 – Combine traditional machine learning features with the generated graph embeddings to form a hybrid feature set.

Step 8 – Select important features using techniques such as Mutual Information and XGBoost feature importance.

Step 9 – Train machine learning classifiers such as Random Forest using the selected features.

Step 10 – Apply the trained model to classify accounts as bot or human using probability thresholds.

Step 11 – Apply the LIME explainability method to identify the most influential features ,along with LLM based summarization.

Step 12 – Stop

#### A. System in Action

The process starts by cleaning and preparing the data, which includes fixing errors and organizing the information. The system then constructs a similarity-based network of users and generates graph embeddings using GraphSAGE, enabling it to capture structural relationships between accounts. Using a combination of data about individual users and these graph-based representations, the system predicts whether an account is a bot, employing ensemble model like Random Forest, and it also provides explanations for its decisions, highlighting the key factors that led to the bot detection.

#### B. Preparing the Data

The system cleans up the user data by handling missing information and inconsistencies, and it calculates various metrics, such as how often a user posts, how quickly they gain followers, and how users interact with each other, while also analyzing the content of their posts.

#### C. Building the Relationship Map

The system creates a network of users based on how similar they are to each other and uses a technique called GraphSAGE to create a representation of each user within the network, capturing both their individual behavior and their relationships with others, which is crucial for spotting coordinated bot activity.

#### D. Making the Prediction

The system uses the prepared data and the relationship map to determine whether an account is a bot, with the Random Forest classifier analysing both individual user

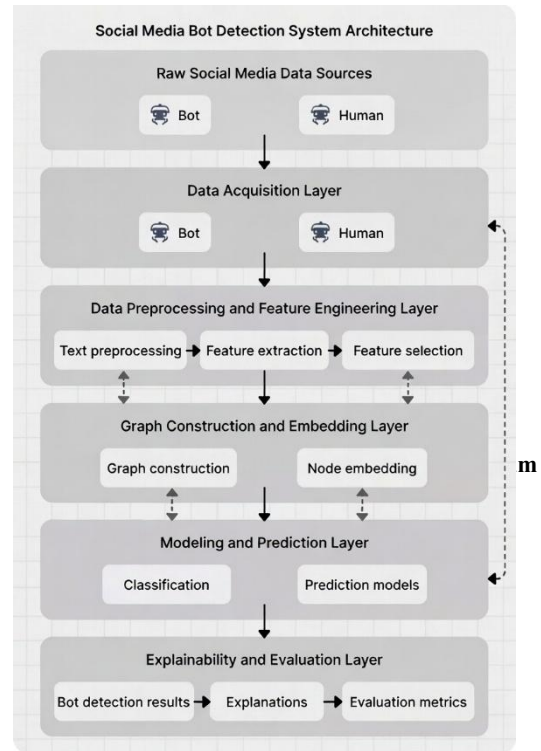
features and graph-based relationship features to produce a  $P(\text{Bot})$  probability score. A classification threshold of 0.39, derived empirically through precision-recall analysis on the training distribution, is applied to assign each account a final BOT or HUMAN label, ensuring reliable detection even when bot accounts are significantly fewer than genuine users.

### E. Explaining the Decision

The system provides interpretable insights into why an account was flagged as a bot, with tools like LIME highlighting the most important features contributing to the classification, such as posting frequency, follower spikes, engagement patterns, and content behavior, providing transparency that allows researchers and platform moderators to understand, trust, and investigate the detection results effectively. This lime result is then passed to LLM to generate a overall summary.

## IV. SYSTEM ARCHITECTURE

The proposed bot detection system is designed in five stages to turn raw Twitter information into reliable bot identifications. The Data Acquisition Layer gathers information from sources like user profiles, activity records, and tweet content, pulling data from Cresci-17 and TwiBot-22 datasets and ensuring consistency and proper formatting. The Data Preprocessing and Feature Engineering Layer prepares the data by cleaning, handling missing values, adjusting numerical values, and creating new features such as tweets per day, sudden follower increases, and sentiment, covering both structured and unstructured data. The Graph Construction and Embedding Layer builds user similarity graphs and creates embeddings that capture individual traits and network influence. The Modeling and Prediction Layer performs bot detection using predictive models, handling class imbalance, calibrating probabilities, and setting optimal thresholds to detect both individual and coordinated bots. The Explainability and Evaluation Layer ensures transparency by highlighting important features, and evaluates performance with metrics like accuracy, precision, recall, F1-score, and ROC-AUC, supporting continuous improvement and real-time monitoring. This system provides a flexible and scalable framework for detecting Twitter bots.



This above figure shows the architecture of a Twitter bot detection system, illustrating a layered pipeline starting from raw Twitter data sources.

### A. Architectural Overview

At a high level, the architecture processes raw Twitter data through sequential analytical stages. Initially, it gathers raw data from Twitter, pulling information from user profiles, activity records, and the content they share. This data is then checked and organized to make sure it's consistent across different sources, like the Cresci-17 and TwiBot-22 datasets. Next, the data gets cleaned up and prepared for analysis. This involves filling in any missing information, adjusting numerical values to a consistent scale, and calculating new metrics. These new metrics include things like how often someone posts, sudden increases in followers, and how positive or negative their posts are. After that, the system creates a visual representation of the data using graphs. These graphs show how similar users are to each other and capture both individual user behavior and how users connect within the network. These graph-based representations, along with the original data, are then used to train predictive models to determine if a Twitter account is a bot. The system also fine-tunes the models to make sure they're accurate, even when dealing with bots that are rare or working together. Finally, the system explains its decisions and evaluates its performance. It uses tools to understand which factors are most important in identifying bots and highlights key connections within the network. It also provides detailed performance metrics, such as accuracy, precision, recall, F1-score, and ROC-AUC. In essence, this system is designed to be reliable, able to handle large amounts of data, and easy to understand.

### B. Data Acquisition Layer

First, the system gathers information. It's like collecting all the puzzle pieces – user profiles, what they've been up

to, and what they've written. This raw data forms the foundation for further processing.

### C. Data Preprocessing Layer

The collected data then gets cleaned up and organized. Think of it as sorting the puzzle pieces and getting rid of any duplicates or irrelevant bits. The system also looks for missing values, noisy data, including activity metrics, linguistic signals, and content characteristics, preparing a rich feature set for modelling.

### D. Graph Construction Layer

Next, the system builds a network of users, like connecting the puzzle pieces to see how they fit together. Using this similarity-based graph, it learns how users are structurally related within the network. This helps spot coordinated or sneaky bot activity that might be hard to see otherwise.

### E. Modeling and Prediction Layer

Then, all this information is fed into powerful models that are good at making predictions. These models analyze the data and decide whether an account is likely a bot. Ensemble approaches like Random Forest combine individual and relational features to ensure robust classification.

### F. Explainability and Evaluation Layer

Finally, the system explains its decisions and checks its work. It uses analytical methods and model interpretation techniques to understand why certain accounts were flagged, highlighting the most influential features and behavioral patterns behind each prediction. In addition, a Large Language Model (LLM) is integrated to generate human-readable explanations, translating complex model outputs into clear and structured reasoning that can be easily understood by users or analysts. The system also continuously evaluates its performance using appropriate metrics to ensure consistency, accuracy, and reliability. This layered approach ensures that the results are transparent, trustworthy, and supported by both statistical evidence and contextual explanation. The entire system is designed to be adaptive, capable of scaling to large datasets, and flexible.

## V. RESULTS AND DISCUSSION

The system was evaluated to spot Twitter bots was put to the test. The goal was to see how well it could accurately identify bots, explain its reasoning, and hold up when faced with different types of data. The evaluation looked at hard numbers that is, performance metrics, how easy it was to understand the system's decisions, and how stable it remained when the proportion of bots to real users changed.

| Feature         | BOT | HUMAN |
|-----------------|-----|-------|
| ff_ratio        | 3   | 3     |
| followers_count | 2   | 4     |
| avg_polarity    | 2   | 1     |
| avg_retweet     | 1   | 1     |

|                      |   |   |
|----------------------|---|---|
| listed_count         | 2 | 3 |
| tweet_count          | 1 | 1 |
| unique_word_count    | 1 | 2 |
| url_count            | 1 | 0 |
| avg_mention          | 1 | 3 |
| hashtag_count        | 1 | 0 |
| has_profile_photo    | 0 | 1 |
| profile_completeness | 0 | 1 |
| avg_subjectivity     | 0 | 1 |
| mention_count        | 0 | 1 |
| has_profile_url      | 0 | 1 |
| avg_url              | 0 | 1 |
| has_description      | 0 | 1 |

The above table shows the features that contribute to each class.

#### Bot pattern:

Features that strongly contribute to bot activity are ff\_ratio, followers\_count, activity\_features like tweets, hashtag, urls.

Bot show imbalanced follower – following ratio + high activity

#### Human pattern:

Features that strongly contribute to human activity are has\_profile\_photo, profile\_completeness, has\_description.

Human shows complete and natural profiles.

While some features like followers\_count and ff\_ratio appear in both classes, their importance and combination with other features differ significantly, enabling the model to distinguish between bots and humans.

## VI. CONCLUSION

This work presents a robust and practical approach for detecting Twitter bots by combining behavioural feature engineering with graph-based representations derived from GraphSAGE embeddings. By integrating user profile attributes, activity patterns, and relational network information, the system effectively captures both individual anomalies and coordinated bot behavior. Evaluated on benchmark datasets such as TwiBot-22 and Cresci-17, the model achieves strong performance, with high accuracy and AUC, supported by careful preprocessing, feature selection, and balanced training using a Random Forest classifier. The hybrid design is a key strength, as it allows the model to go beyond isolated signals and leverage structural relationships between users, making it both reliable and adaptable for real-world deployment.

While the system performs well, there are several opportunities for improvement. One important direction is enabling real-time detection by integrating the model with twitter APIs, allowing continuous monitoring and faster response to emerging bot activity. Additionally,

incorporating adaptive learning mechanisms would help the model evolve alongside changing bot behaviors without requiring frequent manual updates. Expanding the feature space to include multimedia data and richer metadata could further enhance detection accuracy. Future enhancements may also involve deeper use of advanced AI models, such as extending LLM capabilities for content-level analysis and improving interpretability through techniques like attention mechanisms. Finally, extending the framework to other platforms like Instagram and Reddit would increase its applicability while leveraging the existing modular design.

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